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Behaviours and practices in socio-technical systems and sustainability transitions

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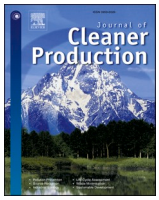
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Editorial



Behaviours and practices in socio-technical systems and sustainability transitions: editorial to the virtual special issue

A B S T R A C T

This virtual special issue relates to the challenge of connecting (a) approaches to understanding individual behaviour, with (b) the macro-level perspectives of 'system' used in sustainability transitions studies. Sustainability transitions studies has traditionally been concerned with the larger picture of technological change in relation to societal, economic and institutional change. Yet increasingly, some authors have argued that neglecting the micro-level details of human behaviour, practice and decision-making means that important processes are being obscured. In this editorial, we illustrate approaches to studying different types of individual behaviour within sustainability transitions frames. The papers in the VSI firstly illustrate the range of human behaviour relevant to transitions, which we categorise as reflective, automatic, strategic and/or everyday. We then further categorise the VSI papers in terms of the strategies that they use for making multiscalar connections. These options include: (i) Bridging concepts: concepts that are inherently both social and individual; (ii) Integrative perspectives: theoretical or conceptual approaches that inherently connect individual and social processes by design; (iii) Multiscale linkages: multi-scale process-tracing within a sociotechnical change process; (iv) Multi-lens or parallel linkages: multiple theoretical lenses and/or methods used in series or parallel. Taken together, the papers in the VSI illustrate the use of these approaches, as they seek to make connections between individual behaviour and the wider sociotechnical contexts in which this takes place.

1. Introduction: why a special issue at the intersection of behaviour and practices in socio-technical systems and sustainability transitions?

Sustainability transitions are long-term, multi-dimensional, and fundamental transformations through which established socio-technical systems shift to more sustainable modes of production and consumption (Markard et al., 2012). It is well established and recognised in the field that understanding of the actions, agency, practices, and behaviours of actors, as well as their enablers or barriers, are critical to the dynamics and governance of sustainability transitions (Köhler et al., 2019). Responding to early critiques of lacking sufficient attention for actors, agency, power and politics, traditionally, sustainability transitions research has relied strongly on disciplines and approaches from the structural and relational branch of social sciences, to engage with actor-oriented research questions and themes. For instance, by combining, integrating or confronting traditional frameworks and concepts from transition studies such as the multi-level perspective (Geels, 2002), with neo-institutional theory (Fuenfschilling and Truffer, 2016; Geels, 2004), evolutionary economics and STS (Geels, 2020), social practice theory (Hargreaves et al., 2013), structural approaches to power (Avelino and Rotmans, 2009) or actor-network theory (Jørgensen, 2012), a rich and nuanced body of work and understanding has developed that analyses the interplay between actors and their structural and relational environments, and how this interplay enables or stalls transition dynamics such as niche experimentation and regime transformation.

What has received considerably less attention is research that draws on disciplines and approaches that focus on individual and actor-internal factors and processes and their role in, and interplay with, change and transformation to achieve sustainability outcomes at the level of sociotechnical systems. Research into everyday behaviours into sustainability transitions has predominantly taken a social practice approach (Kaufman et al., 2021). In contrast, in the context of wider energy social science research, much more pluralism is found in terms of engaging with behaviour and practice approaches, ontologies, and epistemologies (Huttunen et al., 2021). Despite this, energy social science research has - arguably - still not yet made full use of the breadth and depth of insights on behaviour and practices in ways that are set in their in sociotechnical contexts - i.e. in ways that draw on the insights available in both the sustainability transitions literatures *and* the behaviour and practice literatures together.

Approaches embedded in disciplines such as social psychology and behavioural economics have long engaged with the study of what drives, enables, and constrains individuals. More recently, these insights and approaches, under the academic heading of 'behavioural science' and the policy-oriented heading of 'behavioural public policy', are growing in visibility and influence, as they are put to the task of understanding, explaining, enabling, and achieving public outcomes in areas such as health, education, criminal justice, and sustainability (Hallsworth, 2023; Kaufman et al., 2021). Not without critique of, for instance, limited impact, mechanistic thinking, neglect of social context or ethical

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concerns, behavioural science in general "uses scientific methods to generate and test theories that explain and predict the behaviour of individuals, groups and populations" (Hallsworth, 2023).

This VSI is not the first to observe the above gaps in the sustainability transitions literature: an emerging and fascinating debate, and calls for further research, have emerged on advancing understanding between the nexus of behaviour/practices and sustainability transitions – whether from a behaviour science or sociological perspectives (Bögel and Upham, 2018; Huttunen et al., 2021; Kaufman et al., 2021; Upham et al., 2020). Engaging with these emerging debates requires acknowledgement of long-standing epistemological and ontological schisms regarding how to understand, conceptualise and investigate individual behaviours and practices, their drivers and barriers, their institutionalisation and their role in reshaping systems and structures. Behaviour itself, as a scholarly concept, is not an uncontested notion, and some scholarly work embedded within the social sciences has preferred to use other terminology, such as practices, on the basis of opposing ontological, epistemological or ethical grounds, in particular when exploring what people do and why in the context of everyday life (Shove, 2003), including how this can or should inform sustainability policy and governance (Shove and Walker, 2007). Acknowledging such long-standing scholarly debates, and that any scholarly language will carry with it a particular history, here we use both terms - behaviour and practice - to simply coin the empirical phenomenon of the way that a person acts in a particular way in a particular situation or under particular conditions, and to consider the various ontological and epistemological approaches to observe, analyse, or theorise this phenomenon.

Such challenges posed by spanning scales, epistemologies and ontologies of analysis are also far from new, both per se in sociology (Ritzer, 2008; Raub et al., 2011), and in the context of sustainability transitions theory-development. Well over a decade ago, Geels (2010) identified four approaches to this: (i) complete theoretical integration, which often isn't possible; (ii) acknowledging incommensurability; (iii) pragmatic eclecticism; and (iv) inter-ontology crossover, which aims for inter-play where possible. Geels (2010) positions the MLP in the latter category, which focuses on bridging theories, mechanisms, and concepts. Our approaches offered here, illustrated by the papers of the VSI, broadly correspond to (iii) and (iv). The VSI is thus premised on the assumption that fundamental differences between the domains and approaches of behavioural science, social theory, socio-technical systems and sustainability transition studies can be selectively bridged and hence provide new and complementary insights into several key research themes pertinent to the study of sustainability transitions. Before outlining the approach of the VSI further, we will elaborate in more detail why we consider this an important step for socio-technical systems research and transition studies and the challenges inherent in this task.

First, it is well accepted that sustainability transitions in general cannot rely only on technological change alone. Changes in behaviour and practice are deeply part of any transition in general (Creutzig et al., 2022) and in specific contexts (e.g. Fricaudet et al., 2023). However, so far the spectrum of exploring behaviour and practice in transitions has remained limited, and existing contributions in the transitions field have prioritised singular approaches, e.g. the theory of planned behaviour, or social practice theory, over seeking complementarities across approaches. The lack of plurality in perspectives potentially limits our understanding of the role of behaviour and practice in sustainability transitions and the ways in which it influences or is shaped by critical sustainability transition processes such as niche experimentation; using, scaling and diffusing sustainable technologies or innovation systems; regime transformation or decline; or understanding the build up of pressures for change emerging from social movements.

Second, engaging with more diverse approaches to behaviour and practice is particularly important at this point in time because transitions in various domains - including, among others, energy transitions - are moving (or have to move) into new, accelerated phases, which are likely to have implications for every aspect of both our personal and professional lives (Markard et al., 2020). Not only does this likely increase the types of dynamics, diversity of contexts and scales at which behaviour/practice matters and change occurs, but it may also lead to fiercer pushbacks and resistance, as transitions move beyond early adopters. There is a need for new conceptual tools and developments to anticipate and respond to these issues, for broadening the conceptual repertoire, and for clarifying when, how and which concepts and approaches are useful for particular types of transition problems – and which types of behaviours and practices. There is also the need to do this without neglecting the structures that shape behaviour (Shove, 2010).

Third, existing approaches to behaviour/practice in transition studies have tended to narrowly focus on 'users' or 'consumers' in the context of consumption or use of new technologies, whilst giving less attention to the actions of 'powerful' actors such as corporate managers, community-leaders or policy makers. This is important, because such actions, in organized contexts, disproportionately impact the opportunities and challenges for achieving transitions, including influencing the conditions that shape the behaviours of others. In addition, powerful actors' decisions may be influenced by (sometimes) narrow perceptions of what drives consumer or citizen behaviours and practices, resulting in simplistic assumptions of how to change them, or misjudging the relative ease or difficulty of adopting a given behaviour/practice, and what types of support may help.

In sum, while the role of actors and their behaviours/practices is axiomatic for sustainability transition scholarship, the roles of actor-internal factors and processes in behaviour and practice and its nexus with system-level transition processes remain nonetheless narrowly explored empirically and theoretically. In other words, transitions studies acknowledges that individual actors are important, but it has neglected fields that study how that behaviour arises. Likewise, while behavioural science has developed a rich portfolio of concepts, methods and frameworks for explaining, predicting and influencing (sustainability) behaviour, it has paid limited attention to 'seeing the system' in which behaviours are situated (Hallsworth, 2023); whereas arguably social theory applied to the study of practicing provides a more substantial understanding of the structured nature of individual action, but has been criticized for limited policy influence (Rauschmayer et al., 2015). The result is a lack of practical, conceptual, and more comprehensive and policy relevant understandings of behaviour and practice in socio-technical systems and sustainability transitions overall. Arguably, underpinning these gaps and limited interactions, so far, is a long-standing scholarly debate and (un)willingness to engage with fields that are ontologically and epistemologically different.

2. Categorising types of behaviour and practice

One way of characterising the study of actors in sustainability transitions has been introduced by Kaufman et al. (2021) in a review of behaviour in sustainability transitions. In our overview of the VSI papers below, we classify each paper in terms of which type of behaviour is predominantly analysed. Kaufman et al. (2021) distinguished between four perspectives on behaviour in sustainability transitions: reflective behaviour, automatic behaviour, strategic behaviour, and everyday behaviour. In short, *reflective behaviour* conceptualises behaviour as reflective, effortful and conscious: it is frequently analysed as one-off, discrete and observable actions, especially by individuals. These behaviours are causally explained by identifiable phenomena such as psychological and structural "drivers" or "barriers". Behaviours are typically seen as boundedly rational for the individual but may ultimately be variously adaptive (i.e. beneficial) or maladaptive for the individual and/or their community, depending on the scale and nature of analysis (e.g. spatial, temporal, ecological, economic, or social).

Automatic behaviour conceptualises behaviour as often unconscious, intuitive, easy and adaptive (but difficult to change once embedded). It

is usually analysed as repeated, discrete and observable actions, especially by individuals, often referred to as habits. Automatic behaviours are typically maintained by recurring cues or constraints within an individuals' psychological or structural context. From a policy and intervention perspective, behaviour change results from both designed/intentional and unintentional changes in these cues or constraints, for example by governments or businesses manipulating the design and prompts of high-volume service interactions online, in supermarkets, finance, via smart meters, and in forms/letters. Actors are understood to be only able to reflect on and self-elect to change habits when they are disrupted, for example by a substantial change in context.

Strategic behaviour conceptualises behaviours as strategic, intentional actions undertaken for comparative benefit in a competitive context, especially by niche innovators, established firms and government agencies and other organisational actors. Expectations of behaviour, and behaviour change, of others can be a focus – for example beliefs and assumptions about 'what works' within particular institutional and material (socio-technical) contexts to change others' behaviour matters, and informs strategy by firms, policy makers, NGOs and community groups. Strategic behaviours seek to advance an actor's interests by influencing the individual or collective behaviour of others or turn others' influencing attempts to the actors' advantage.

Finally, Kaufman et al. (2021) define the concept of *everyday behaviour* as resisting reduction of the analysis to the scale of individual behaviour, and instead conceptualising behaviour as clusters of actions (practices) that emerge from and are reinforced by relationships between individuals in context. While social practices theories are prominent in this form of analysis, we see this perspective as potentially including other approaches with relatively 'bottom up' and 'flatter' ontologies, plus more relational and interpretivist approaches.

3. Approaches to studying individual actors in sustainability transitions frames

In this section, we describe the main research approaches used to study individuals within sociotechnical sustainability transitions frames, and which in section 4 we show to be variously applied in the papers of the VSI. For each approach, there is flexibility in methods of data collection and analysis, which may be qualitative, quantitative or both. Likewise, transdisciplinary approaches play a growing role in this strand of research (Bögel et al., 2023a; Trenks and Bögel, 2024).

The first research approach consist of using *bridging concepts*, by which we mean concepts that refer to processes or phenomena located or held at both individual and pan-individual (social) levels. The premise is that, by virtue of their nature, bridging concepts can frame, characterize and aid understanding of transition processes in which individuals play a direct role. Examples of bridging concepts used for this purpose include social representations, sociotechnical expectations, narratives and other broad terms that can be applied to the study of phenomena that take place at different scales - such as general sources of conflict. Each such concept allows for individual difference, itself in part related to the social and sociotechnical positions of particular individuals, such that the social and individual are interrelated. Examples include the concept of social representations, which concern the ideas with which people view the world (Upham et al., 2015; Giardullo et al., 2019); sociotechnical expectations, which concern how people expect the world to be (Levidow and Upham, 2017); narratives, which concern the chronological accounts with which people explain situations (Upham and Gathen, 2021; Dobroć et al., 2023); and conflict, which can be topologized in terms of its role in positional difference (Bögel et al., 2024).

The second type of research approach is the use of *integrative perspectives*: these are theoretical or conceptual approaches that inherently connect individual and social processes *by design*, latterly with sustainability transitions theorisation in mind. Hence applications of strong structuration, designed by (Stones, 2005) and applied by (Hermwille, 2016; Upham et al., 2018) in sustainability transitions contexts, is explicitly designed to reflect structure and agency as both internal and external to individuals, and its relevance to sociotechnical analysis was quickly recognised (Greenhalgh and Stones, 2010). Another such framework is the Embedded Agency Perspective (EAP) (Bögel et al., 2022; see also Augenstein et al., 2022), which proposes an integrative socio-spatial framework consisting of four dimensions: physical, cultural, actors and agency, and regulation. These dimensions are used to characterize the interplay of individual lived experience; social (interpersonal) discursive elaboration; and the societal level of symbolically mediated experience (Bögel et al., 2022). The approach was designed for the analysis of scaling spatially-situated sociotechnical experiments (living labs) but has wider applicability.

A third research approach can be described as *multiscale linkages*, which consists of multi-scale process-tracing within a given sociotechnical change process connecting individual and system or larger structural levels. These linkages may be observed post-hoc or longitudinally. Hence (Bögel et al., 2023b) focus on the role of identity to explain participation in energy communities (Becker et al., 2021); focus on the role of changing social identity in institutional work relating to mobility transitions (Bögel et al., 2019); focus on connections between individual, organizational and multi-organisation processes in the context of regional sustainability transitions; and Raven et al. (2025) focus on connections between household practices and wider socio-technical systems in the case of low waste living. Many papers linking scales of analysis use this approach to some extent.

Recent work has also begun to combine approaches two and three: Trenks and Bögel (2024), using the Embedded Agency Perspective (Bögel et al., 2022), show how individual concepts are influenced by, but also influence, the socio-spatial structure. For example, they show in one case how, at the start of a sustainability transitions experiment, legal regulations can be a barrier to citizens' self-efficacy. Yet, through the building of a strong group identity in the context of a sustainability transitions experiment (Bögel et al., 2023b), this barrier can be turned into a key motivation for citizens to foster structural changes in the system, which, in turn, enhances their perceived self-efficacy. Further research could deepen this approach by connecting to other bridging concepts and/or using it to trace multi-scale linkages, e.g. identity development, embedded in its socio-spatial context.

The fourth approach, which we call *multi-lens or parallel analysis*, connects different paradigms or theoretical lenses in series or parallel, to shed different types of light on a problem (e.g. see Geels, 2009). This may also be used as part of the third approach of multi-scale path or process tracing or process. To some extent this type of approach is already used in multi-disciplinary or interdisciplinary work, but here we mean the integration to be tighter and - emphatically - the purpose is to contribute to an understanding of sociotechnical processes, not to deepen knowledge in individual disciplines.

4. Overview of the VSI papers

In this section we provide an overview of the papers in the VSI, describe how they relate to the four different approaches for studying behaviour in sustainability transitions, in terms of which they use predominantly (though often not solely), and classify them in terms of the four types of behaviour or practice studied. Table 1 provides an initial overview, followed by narrative summaries of the papers.

Guibentif and Patel bring together agency theories - i.e. theories of how people can change the social and sociotechnical structures in which they live, and also how those structures shape and constrain people - with discursive approaches, to study transition intermediaries. 'Discursive approaches' here means that the author studies what people say, to infer their implicit beliefs and understandings. In terms of the four approaches above, the paper traces multi-scale linkages (research approach 3) between individual cognition, organisational strategy and wider system change via the use of elements of an integrative approach

Table 1
Overview of the VSI papers.

Authors	Concepts/frameworks	Empirical domain	Geographical focus	Methods	Research approach (see section 3: bridging concepts; integrated frameworks; multi-scale linkages; multi-lens or parallel analysis)	Perspective on behaviour (Kaufman et al., 2021: reflective; automatic; strategic; everyday)
Guibentif and Patel	Intermediation, discourse, strong structuration	Energy	Switzerland	Workshops & interviews, case study	Integrated framework, multi-scale linkages	Strategic, Reflective, (somewhat Automatic)
Wehden et al.	Pro-environmental behaviour, Norms	Energy Workforce	Germany	Mixed method	Multi-scale linkages	Reflective, Automatic
Haaja and Harikkala-Laihinen	Emotions	Manufacturing	Finland	Interviews	Multi-scale linkages	Automatic (elements of Strategic, Reflective)
Schünemann et al.	Modelling approaches	Urban design	Generic	Agent based modelling, system dynamics, causal loop modelling	Integrated frameworks, multi-scale linkages	Reflective, Automatic, Strategic, Everyday
Middha	Science & technology studies	Waste recycling	Australia	Interviews, document analysis & observation	Bridging Integrative	Everyday, Reflective
Tolonen	Organisational studies	Stakeholder management	Multinational	Document analysis	Multi-scale linkages	Strategic
Ertelt et al.	Policy mix perceptions	Heavy Transport	Sweden	Interviews, Surveys	Integrative	Reflective, Strategic
Allert and Rees	Social identity model	Mobility politics	Germany	Theoretical paper	Multi-scale linkage	Everyday and Reflective
Götting & Becker	Political engagement, civic voluntarism, Theory of Planned Behaviour	Mobility politics	Germany	Survey	Multi-scale linkages	Reflective, Strategic
Suski et al.	Practice theory	Pro-sustainability Behaviour	Germany	Interviews; LCA	Bridging; Multi-scale linkages	Everyday
van den Broek et al.	Mental modelling	Energy	Netherlands	Mental model mapping	Bridging	Strategic, Reflective
ten Berge	Mnemonic agency (memory); Identity	Energy	Netherlands	Historical text analysis	Bridging	Strategic and everyday
Long et al.	Multi-level perspective, energy citizenship, game playing	Energy	Netherlands	Interviews, case study	Integrated framework	Strategic
Vasseur et al.	Social practice theory, capabilities approach	Energy	Netherlands	Living lab, survey, longitudinal	Integrated framework, multi-lens/ parallel analysis	Everyday

(research approach 2) - namely Stone's strong structuration (2015). In terms of the four behaviour types of Kaufman et al. (2021), the behaviours in question are primarily reflective and strategic, although some heuristic, less conscious processes are also (almost inevitably) operative. The empirical context is a transition intermediary, i.e. an organisation that brokers knowledge and resources to support other organisations in their efforts towards sustainability.

Wehden et al examine agency-structure interactions in a multi-scale study of young people's navigation of career choices in transitions (research approach 3). They point out that skilled worker shortages in vocational trades, such as photovoltaic installation, pose significant barriers to low-carbon transitions globally. Despite the critical role of work in these transitions, research in this area remains limited. The paper examines factors influencing career choices in Germany's photovoltaic installation sector using case studies and a survey of adolescents. The study highlights the pivotal role of vocational work in sustainable energy transitions and the need for greater recognition and research at this intersection. It adopts a "reflective" perspective on behaviour, while also recognising the chilling effect of more "automatic" pressures such as systemic gender norms and discrimination. These insights, they argue, can and should inform "strategic" perspective interventions by intermediaries, government and business actors for transitions.

Haaja and Harikkala-Laihinen explore emotions experienced in the context of a European manufacturing industry undergoing a green transition. On the basis of interviews therein, they formulate three propositions at different stages of the emotion elicitation process, reflecting actors' interpretations and agency. Firstly, they propose interrelationships between triggers (e.g. conflict, changes in environment) in transitions processes and emotions (e.g. anger, joy, disgust). Secondly, they propose that stronger emotional responses are related to expectations, especially those of actors perceived to have agency in shaping the transition. Thirdly, in considering the interactions between motivational outcomes and emotion, they propose that actors' beliefs about their own agency in progressing the sustainability transition play a key role in determining the motivational consequences of emotions. By uncovering the emotions experienced by different actors, along with their objects, this study sheds light on the systemwide complexity of advancing sustainability collectively. While predominantly indicating an "automatic" perspective on behaviour in transitions (emotions), it also suggests "reflective" consequences for motivation, which are important dimensions to incorporate in "strategic" perspectives on transitions. The paper uses research approach 3, multi-scale linkages, in that it shows the connections between individual emotions among actors in positions of sociotechnical relevance.

Schünemann et al discuss the merits of alternative approaches to modelling social behaviour in socio-technical systems, for the purpose of assessing policy interventions. When used in a stand-alone form, such models are an example of an integrative approach (research approach 2). However, models may also in principle be brought together loosely to inform one another, in which case this would be a case of approach 3, multi-scale linkages. Connected, multiple model studies would be approach 4. The perspective on behaviour being modelled can in principle represent any of the behaviour types defined by Kaufman et al. (2021). At the micro-level, Schünemann et al. argue that agent-based modelling (ABM) can help to explore how individual behaviour can scale or cascade socially and hence affect sociotechnical systems. In contrast, the macro-level approach of System Dynamics models (SDM) allows investigation of how system interactions drive complex system behaviour, but tends to allocate individuals to homogeneous groups. Qualitative modelling approaches such as Systems Thinking Modelling

(STM) with causal loop diagrams informed by expert or participant knowledge is relatively descriptive, less parameterised and less onerous.

Middha investigates waste recycling by people living in Melbourne apartments, which are typically associated with relatively low recycling rates for various reasons, both practical (e.g. hygiene) and governance (e.g. split responsibilities among different agencies). As a behaviour, while much recycling can likely be categorised as everyday or automatic in the terms posed by Kaufman et al. (2021), Middha also shows how some is much more reflective. As an approach to conceptually connecting individuals and structures, social practice theory is integrative (research approach 2), though Middha also uses bridging concepts (research approach 1), in her emphasis on boundaries as entities that materially and conceptually connect people and things. Specifically, Middha uses a boundary object-focused, social practice perspective, to understand the situation and to make recommendations. Overall, the study illustrates the implications of individually and socially held meanings and practices for wider (waste) system functioning.

Tolonen's article 'Keeping up with the Decarbonization' is about organisational, rather than individual behaviour - one of the few of this type in this Special Issue. The paper is of the multi-scale type, research approach 3, and represents a strategic perspective on behaviour. It characterises the reflective organisational strategies that have to date helped airlines avoid large scale use of substitutional technological options arising from sustainability pressures, such as synfuels or biofuels, as well as more substantial system reconfigurations, such as more rapid substitution of flights by rail travel. Tolonen's premise is that in the transitions literature, incumbent firms are too often treated as homogeneous in character and behaviour, whereas in practice their response to sustainability pressures, and hence their effect on transitions processes, are more differentiated. The paper contributes to an understanding of incumbent responses under transition pressures and illustrates the value of connecting the transitions literature with the organisational behaviour literature, specifically corporate sustainability.

Ertelt et al connect individual perspectives, organisational behaviour and the transition to zero emission vehicles (ZEV), investigating the influence of individual managers' policy mix perceptions on the uptake of ZEVs in Sweden. The findings reveal a positive correlation between managers' perceptions of technology neutrality and adoption delay. They suggest that a key mechanism involved is anticipated regret, which mediates the positive correlation between perceived technology neutrality and the ZEV adoption delay decision: the more technology neutral the policy mix is perceived, the more anticipating regret over adopting ZEVs is expressed by the managers, which, in turn, increases the likelihood to delay ZEV adoption. Moreover, managers' perceptions of the rate of technological change affects their anticipated regret. The work contributes to the transitions literature by unveiling how a specific policy mix perception, namely technology neutrality, could delay the adoption of ZEVs, calling for a nuanced re-evaluation of the current policy mix to facilitate a faster transition towards net-zero goals. It also underscores the importance of paying closer attention to how individual actor-internal factors (such as perceptions and emotions) of individual decision-makers in organisations may influence transition processes. The research succeeds in bridging both reflective and strategic perspectives on behaviour in its use of the multi-scale approach (3).

Allert and Reese examine citizen engagement in sustainable mobility in relation to structural challenges. Drawing empirically on examples from the German context, the authors use a socio-psychological theory of collective action, namely the social Identity Model of Pro-Environmental Action (Fritsche et al., 2017), to highlight how potential motivational factors such as a common social identity, collective efficacy beliefs and ingroup norms and goals are difficult for citizens groups to leverage, due to path dependencies within the mobility regime. They set out an interdisciplinary research agenda accordingly, focused on understanding citizen engagement in mobility activism, addressing the interconnections of social psychology, collective activism and sustainability transitions. The authors' research approach is multi-scale (approach 3), where psychological and structural phenomena connect scales of analysis and different types of phenomena. The study similarly connects reflective and strategic types of behaviour.

Goetting and Becker present an empirically driven analysis integrating political science and psychological theoretical explanations for levels of political engagement, with the transition away from land consumption by public parking spaces. Each car park removed opens up space for cycling infrastructure, non-commercial public spaces, and urban greenery. However, car use is the dominant mode and such urban transitions are often accompanied by participatory processes or political actions, such as demonstrations or signing petitions, with those with something to lose (drivers) often much more engaged than those with something to gain (pedestrians, and active, shared and public transport users), who may also have fewer resources or capacity to engage. Through its multi-scale approach (3), the study highlights the importance of psychological variables such as perceived behavioural control, alongside structural factors. The behaviour types involved are both reflective and strategic.

Suski et al. highlight the role of social innovations, bottom-up initiatives and new narratives in the development of new social practices for a circular economy, combining this argument with environmental assessment of such bottom-up initiatives and emerging narratives. The paper connects the study of (changed) practices and behaviours to the socio-structural foundations of transition theory (Köhler et al., 2019). Social practice theory is here used as both a bridging concept (research approach 1), also aligning with the everyday perspective on behaviour in Kaufman et al. (2021). In addition, the study offers first ideas on multi-scale linkages (research approach 3), as it aims to answer the question: How do changes in social practices on the actor level influence the system-level? The paper offers not only a conceptual approach but also uses an illustrative case study of a bottom-up neighbourhood initiative.

van den Broek et al (2024) utilize the concept of mental models to explore varying understandings of the drivers behind the energy transition among citizens and experts in Norway and the Netherlands. The authors apply the M-Tool, an innovative online tool that allows participants to visualize their assumed relations of drivers of the energy transition. As such, the authors provide insights into how cognitive representations differ between citizens and experts as well as between citizens from different regional contexts. Additionally, they show how individual characteristics such as political orientation and climate change worry relate to the mental models. By investigating the mental models of actors across a sociotechnical system, the paper illustrates how a psychological approach can reveal how actors see themselves embedded within a system and how they view potential drivers of change. For this purpose, the concept of mental models functions as a bridging concept (research approach 1) across levels of analysis, by virtue of these being viewed as simultaneously individual and social. The paper adopts a reflective perspective on behaviour, but in engaging with explanations for transitions dynamics it also indicates a strategic perspective.

Ten Berge examines the role of mnemonic agency in energy transitions, investigating collective and individual memories of energy systems, and the interplay of women's identities in relation to the historical

Dutch transition to gas and electricity. The author draws on texts from the time, observing how these use process of "mnemonic imagination" to craft cohesive narratives that reinforced collective identities (including gender-stereotypical norms). For example, ten Berge shows how the political views of the authors of guides for women's behaviour influenced their accounts of history's trajectory, in turn shaping their visions of what was possible in the future. The article concludes that during this historic energy transition, the agency of individual authors, women's organisations, and political movements functioned in an interconnected and layered manner, collectively shaping historical discourse. Identity is again used as a concept that bridges the individual and the collective (research approach 1), the social and technical, mediated by memory, used strategically but in relation to everyday behaviour. The study has implications for the study of implicit norms in sociotechnical futures, and how these shape behaviour, which in turn shapes technology.

Long et al extend Geels (2002) Multi-Level Perspective with insights from the literature on energy citizenship and game theory. The study investigates three propositions: 1) that energy citizens will respond to landscape dynamics by re-orienting action towards a purpose or goal centred around renewable energy; 2) that citizens will adjust new personal or collective renewable energy goals by aligning or adapting cognitive and normative rules; and 3) that energy citizens will use rule-bending to change the terms of their enrolment in the production processes of energy market regimes and in governance processes in energy policy regimes. On the basis of interviews with residents living close to wind turbines, the authors find that individuals shift goals in response to landscape pressures, and that their perceptions of these pressures can influence the cognitive-pragmatic and normative rules that they use, and how they relate to regulative rules. The paper is an example of conceptualising behaviour through a strategic lens. Arguably, the predominant research approach in this paper is research approach 3, multi-scale linkage, but there is also the use of two theoretical lenses (approach 4).

Vasseur et al combine social practice and capability theories to investigate socio-cultural factors relevant to home energy use, in pursuit of a substantial reduction in energy consumption. Data collection is a combination of living lab interventions and a follow up study using survey methods to investigate the long-term effects on practices of both the living lab intervention, and of broader landscape disruptions. A key finding is that the living lab interventions enabled participants to gain new capabilities, which made them better equipped to respond to the disruptive events of sharply rising energy prices and Covid-19. The study focuses on everyday behaviour, and the combination of social practice theory and the capabilities approach exemplifies how the development of an integrated framework (research approach 2) is useful for studying individual actors in sustainability transitions. The study also has some features of the multi-lens or 'parallel analysis' research approach (4), as the use of both frameworks to study the same empirical phenomenon allows the researchers to move back and forth between the two, enabling for a more comprehensive and reflexive interpretation of the data.

5. Conclusions and research directions

A wealth of theories and perspectives on human behaviour have been developed, ranging across multiple fields. As the field of sustainability transitions broadens, it is also beginning to become more multi-scalar in its concepts and theorisation. An interdisciplinary field from the beginning, more attention is being given to the role of micro-level phenomena as contributory to meso and macro scale change. The challenge in connecting perspectives is to retain the value of each at their respective scale of analysis. Reduction of individual behaviour to aggregated outcomes has the value of simplification, but fuller comprehension requires understanding details that are elided in general frameworks.

Across the 14 papers of the VSI, we show how four different research approaches can be used to connect individual-level analyses with sociotechnical sustainability transitions concepts. These are not mutually exclusive in theory or practice, but they nonetheless differ. Often, authors do not seek a close theoretical coupling between scales of analysis, but rather an empirical one. This has the value of being immediately, practically relevant, but we would also like to see more theoretically coupled work. In terms of further research, there are a wide range of possibilities relevant to the roles of actors in sociotechnical processes. These are likely to involve application of the thinking set out above (four behaviour types and four means of connecting these to transitions concepts), and some are already being pursued. For example, the emphasis on norms in institutional theory (Geels, 2014; Scott, 2008) can be viewed as bridging and integrative. Norm-focused study allows individually-held, internalised norms to be connected to theorisation of the role of social norms as institutional context (Johansen and Upham, 2025). A similar logic applies to individual and social identity as referred to above, but in different contexts, including organizational; likewise the role of professional and other epistemic communities in supporting or resisting change. Sometimes institutional and policy entrepreneurs will be important (Battilana et al., 2009; Huitema et al., 2011). There will be connections to be made here between their individual characteristics and their enabling and inhibiting environments, both organizational and structural (Sorrell, 2018; Svensson and Nikoleris, 2018).

Overall, the possibilities are as numerous as the concepts available, and they can be explored in deliberate research designs, beyond the geographies, sectors, and scales studied in this particular VSI. Social practice theory has also been applied to make such connections (Hargreaves et al., 2013; Huttunen et al., 2021), but is still arguably under-used in relation to sustainability transitions frames. The social practice approach is fundamentally sociological, but practices are nonetheless carried by individuals and the concept of habitus specifically is hence arguably both individual and social (Bögel and Upham, 2018). There is scope for examining practices across all contexts (Bögel et al., 2020). Similarly, power and politics, social and political movements are also socially and culturally shared processes with individually internalised dimensions (Schwartz, 2011), including attitudes, values, beliefs, identities and others.

Finally, acknowledging the relevance of individual-level processes also opens the way for use of individual-level analytic methods and connecting the results of these to structural accounts. Within cognitive psychology, these methods may be associated with relatively positivistic epistemologies (Yadegari et al., 2024), or they may involve more constructivist methods (Klapper et al., 2024). Again we would reiterate that we are not advocating the use of psychological concepts or methods per se, but to further sustainability transitions accounts in which the relevance of individual-level processes is more fully acknowledged and accounted for. Individual-level considerations may also inform multi-scale mathematical models of system change, again involving individuals that behave socially, be this in terms of systems dynamic, agent-based or other modelling forms (Hansen et al., 2019).

Overall, we hope that this VSI goes some way to furthering the study of the individual behaviour and practice as part of sustainability transitions perspectives, by setting out and illustrating the research approaches by which connections may be made for different types of behaviour.

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